

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 5 Practical - Second-order Derivatives and Root Finding (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Explain the concept and significance of second-order derivatives in the analysis of functions.

(2) Compute second-order derivatives of functions to classify stationary values as maxima, minima, or saddle points in optimisation problems.

(3) Describe steady-state systems in physical and mathematical modeling.

(4) Apply Newton's Method to find the roots of complex functions in numerical problem-solving.

Exercise 1: Second-order Derivatives

- In the last practical, we found stationary values for the following function:

$$f(x) = \frac{x^3}{3} + x^2 - 15x + 1$$

which were $x = -5$ and $x = 3$.

- Use the second-order derivative to confirm if they are maxima, minima or saddle points.

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Solution 1: Second-order Derivatives

- For the following functions

$$f(x) = \frac{x^3}{3} + x^2 - 15x + 1$$

the first- and second-order derivatives are:

$$\frac{dy}{dx} = x^2 + 2x - 15 \qquad \frac{d^2y}{dx^2} = 2x + 2$$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Solution 1: Second-order Derivatives

- To confirm whether they are maxima or minima, we evaluate the second-order derivative at the stationary values:

$$\frac{d^2y}{dx^2}(3) = 2 \times 3 + 2 = 8$$

$$\frac{d^2y}{dx^2}(-5) = 2 \times (-5) + 2 = -8$$

- As $\frac{d^2y}{dx^2}(3) > 0$ then $x = 3$ is a minimum.
- As $\frac{d^2y}{dx^2}(-5) < 0$ then $x = -5$ is a maximum.

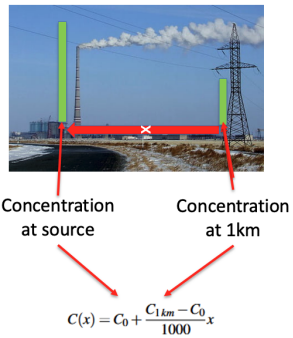
Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
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e^{kx}	ke^{kx}
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Exercise 2: Steady State

- An industrial incinerator produces a steady output of particulate matter and has a concentration of $C_0 = 1\text{mg/m}^3$ of air at the chimney.
- At 1km away from the incinerator plant the concentration is $C_1 = 0.11\text{mg/m}^3$.
- The concentration as a function of distance from the incinerator is given by:

$$C(x) = C_0 + \frac{(C_1 - C_0)}{1000}x$$

- Is this system in steady state?



SOLUTION 2: STEADY STATE

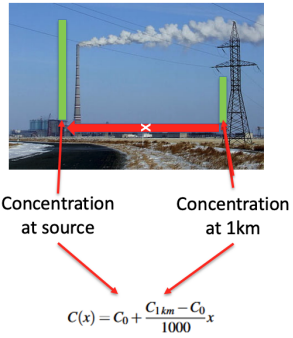
- To deduce if the system is in steady state, we look at the first-order derivative:

$$C(x) = C_0 + \frac{(C_1 - C_0)}{1000}x$$
$$\frac{dC}{dx} = \frac{C_1 - C_0}{1000}$$

- We see that the first-order derivative is constant which means the second-order derivative is zero:

$$\frac{d^2C}{dx^2} = 0$$

- So the system is in steady state.



Exercise 3: Newton's Method for Root Finding

- Find a root of the following equation:

$$f(x) = x^2 + 2x - 10$$

using Newtons Method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- Use an initial guess of $x_0 = 4$ and give the answer to 3 decimal places (i.e. there is no more change in the fourth decimal place).

Solution 3: Newton's Method for Root Finding

- We see that

$$f(x) = x^2 + 2x - 10$$

$$\frac{df}{dx} = 2x + 2$$

- Using Newton's Method with an initial guess of $x_0 = 4$:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 4 - \frac{f(4)}{f'(4)} = 2.6$$

- Continue using Newton's Method:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.6 - \frac{f(2.6)}{f'(2.6)} = 2.32778$$

Solution 3: Newton's Method for Root Finding

- Continue using Newton's Method:

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.32778 - \frac{f(2.32778)}{f'(2.32778)} = 2.31664$$

- Continue using Newton's Method:

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 2.31664 - \frac{f(2.31664)}{f'(2.31664)} = 2.31662$$

- You can see that at the fourth iteration, thus a value for x_4, our value only changes at an accuracy greater than 4 decimal places so we can stop.
- Therefore, we can see the function, $f(x)$, has a root at 2.31662

Solution 3: Newton's Method for Root Finding

- Luckily we can verify this that 2.31662 is a root of:

$$f(x) = x^2 + 2x - 10$$

- Use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times (-10)}}{2} = -1 \pm \frac{1}{2}\sqrt{44}$$

  which means there are roots at $x = 2.31662$ and $x = -4.31662$.

- Newton's Method for Root Finding did find the root correctly!

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